

Midterm Exam: Micro 629 - Fall 2011

Texas A&M University

October 13th, 2011

By writing down your name you pledge to obey TAMU Honor Code:

“An Aggie does not lie, cheat, or steal or tolerate those who do.”

Name (in pen, not pencil): _____

EXAM RULES

- Your interpretation of the questions makes part of the exam. Thus, no questions will be answered during the exam.
- No personal items (bags, cellphone, books, notebooks, calculators, etc.) will be allowed with you during the exam.
- Some of the sections in the different problems require you to prove the preceding sections in the problem; if you cannot prove one, move on, and assume that it is true; use it to prove what is next and then come back if you have time.
- It is valid to use results from class or homework, as long as: (i) I am not asking you to provide an explicit proof, and (ii) you precisely quote the result from class or homework. For instance, it is not valid to write: “by a result from class the proof is completed.” In this case you would have to quote the exact result, with all hypotheses and implications.

	grade
1 (a)	
(b)	
(c)	
(d)	
(e)	
2 (a)	
(b)	
(c)	
3 (a)	
(b)	
(c)	
(d)	
4	
5 (a)	
(b)	
(c)	
Total	

1. (20/100) Let X be a non-empty finite set and B a binary relation on X . Assume that B satisfies the following two properties:

- (a) If $x B y$ then $\neg(y B x)$ (i.e., it is not true that $y B x$.)
- (b) If $\neg(x B y)$ and $\neg(y B z)$, then $\neg(x B z)$.

Let W be the binary relation on X defined by: $x W y$ if $\neg(y B x)$.

- (a) (2/20) State the definition of completeness of W .
 - (b) (5/20) Prove that W is complete.
 - (c) (2/20) State the definition of transitivity of W .
 - (d) (5/20) Prove that W is transitive.
 - (e) (6/20) Prove that there is an element $x^* \in X$ such that for each $y \in X$, $\neg(y B x^*)$.
2. (20/100) Let $(\mathcal{B}, C)$ be a choice structure.
- (a) (5/20) Formally state WARP.
 - (b) (5/20) State Sen's property α and β .
 - (c) (10/20) Prove that WARP implies Sen's property α and β .
3. (20/100) Consider an L -commodity space $\mathbb{R}_+^L$ and a preference $\succsim$ on $\mathbb{R}_+^L$. Assume that the choice correspondence associated with $\succsim$ is single-valued with demand function x . A commodity $l \in \{1, \dots, L\}$ is said to be a luxury for $\succsim$ at (p, w) if the income elasticity of demand for commodity l at (p, w) is greater than or equal to 1.
- (a) (3/20) State the definition of homothetic preferences.
 - (b) (7/20) Suppose that $\succsim$ is homothetic. Calculate $x(p, w)$ as a function of $x(p, 1)$ and w (provide a formal argument). Does $\frac{\partial x_l}{\partial w}(p, w)$ exist for each $l \in \{1, \dots, L\}$ and each (p, w) ?
 - (c) (3/20) Define income elasticity of demand for commodity l .
 - (d) (7/20) Prove that if $\succsim$ is homothetic, then each commodity is a luxury at each (p, w) in which consumption is interior (i.e., $\gg 0$).
4. (20/100) Consider a multi-agent consumption space. Suppose that agents preferences $(\succsim^i)_{i \in N}$ are homothetic and all equal, i.e., for each pair $\{i, j\}$, $\succsim^i = \succsim^j$. Prove that the Aggregate Demand is well defined.
5. (20/100) Let $\succsim$ be a monotone and continuous preference on $\mathbb{R}_+^L$. Let $p \in \mathbb{R}_{++}^L$. Prove that $\succsim$ is represented by the function m_p that associates with each $x \in \mathbb{R}_+^L$ the value

$$m_p(x) \equiv \min_{y \in U(\succsim, x)} p \cdot y.$$

Do it in three steps:

- (a) (2/20) Write the statement " m_p represents $\succsim$ " as an if and only if statement.
- (b) (9/20) Provide an explicit proof of $\Rightarrow$.
- (c) (9/20) Provide an explicit proof of $\Leftarrow$.